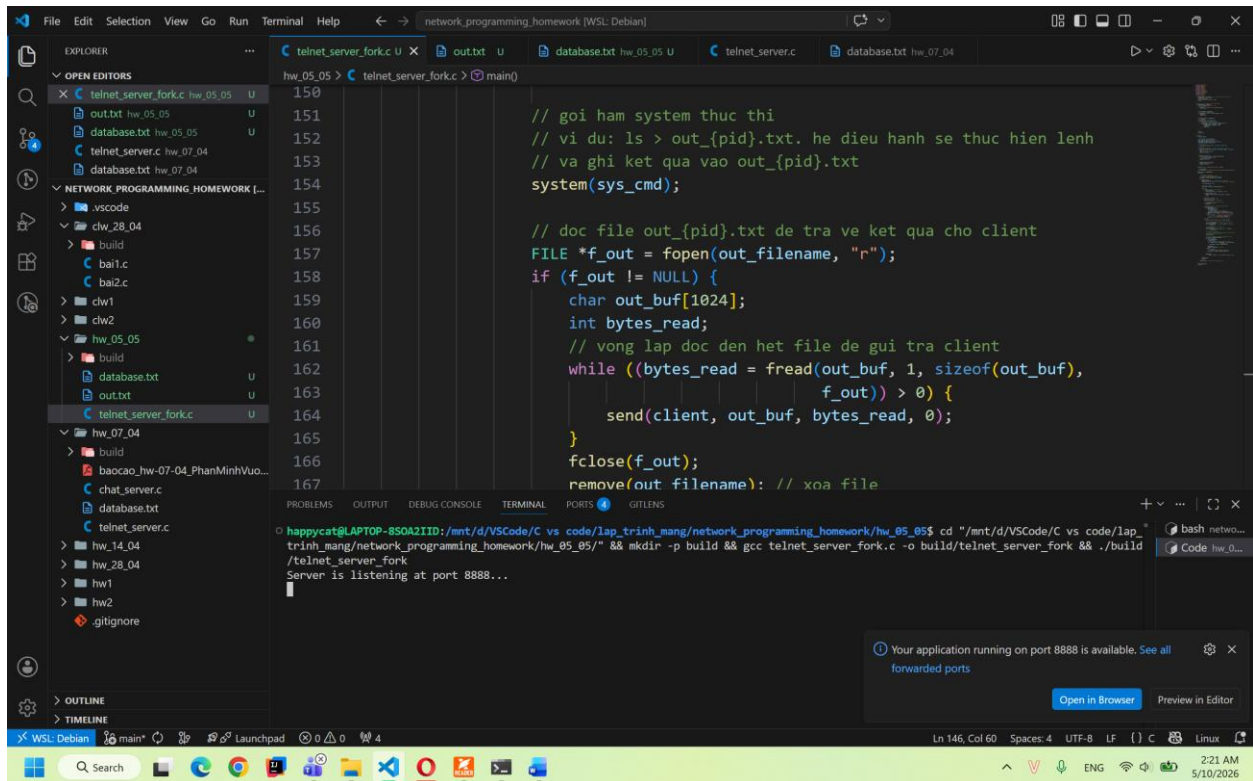


BÁO CÁO BÀI TẬP VỀ NHÀ NGÀY 05/05/2026

HỌ VÀ TÊN: PHAN MINH VƯỢNG

MSSV: 20225241

1. Bài 1 – telnet_server_poll
 - Chạy server ở cổng 8888



The screenshot shows a Visual Studio Code editor window titled "network_programming_homework [WSL: Debian]". The Explorer sidebar on the left shows a project structure with folders like "hw_05_05" and "hw_07_04", and files like "telnet_server.c", "out.txt", and "database.txt". The main editor area displays a C program named "telnet_server_poll.c" with the following code:

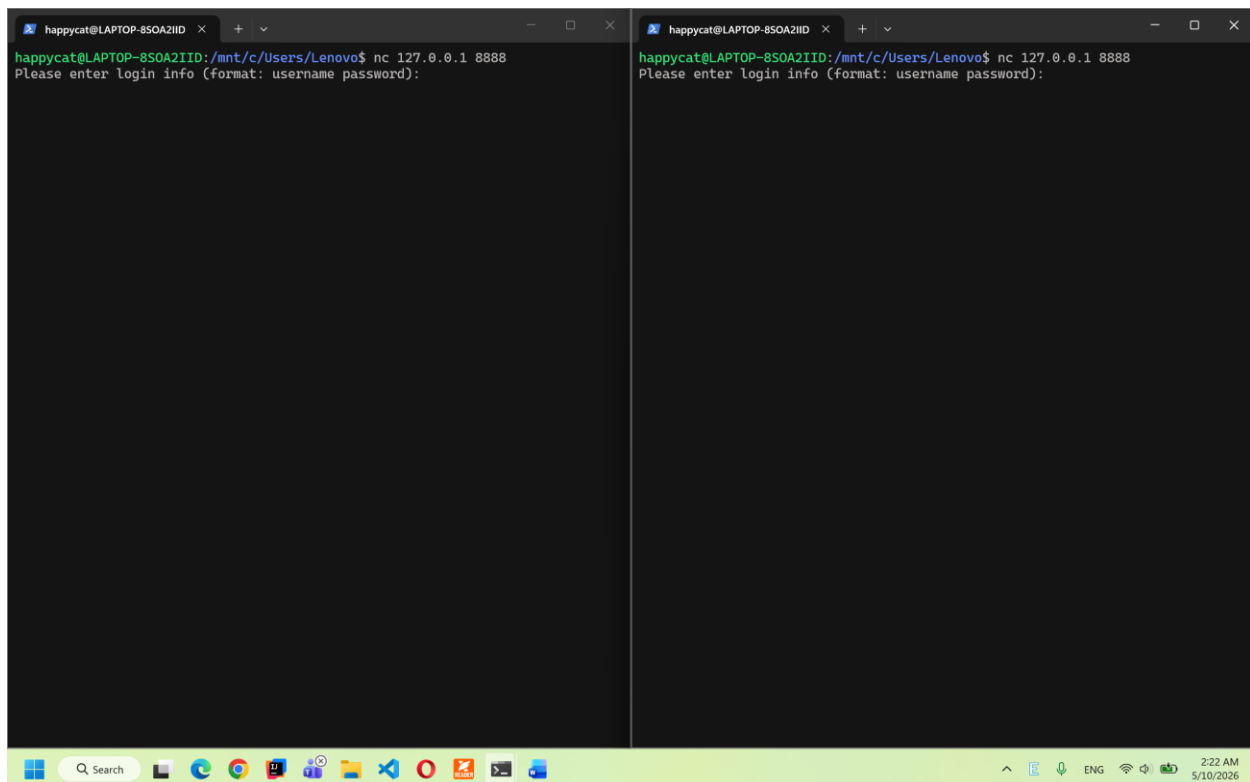
```
150 // gọi ham system thuc thi
151 // vi du: ls > out_{pid}.txt. he dieu hanh se thuc hien lenh
152 // va ghi ket qua vao out_{pid}.txt
153 system(sys_cmd);
154
155
156 // doc file out_{pid}.txt de tra ve ket qua cho client
157 FILE *f_out = fopen(out_filename, "r");
158 if (f_out != NULL) {
159     char out_buf[1024];
160     int bytes_read;
161     // vong lap doc den het file de gui tra client
162     while ((bytes_read = fread(out_buf, 1, sizeof(out_buf),
163                               f_out)) > 0) {
164         send(client, out_buf, bytes_read, 0);
165     }
166     fclose(f_out);
167     remove(out_filename); // xoa file
```

The terminal at the bottom shows the command prompt for "happy cat@LAPTOP-8SOA2IID" and the execution of the program. The output indicates that the server is listening at port 8888:

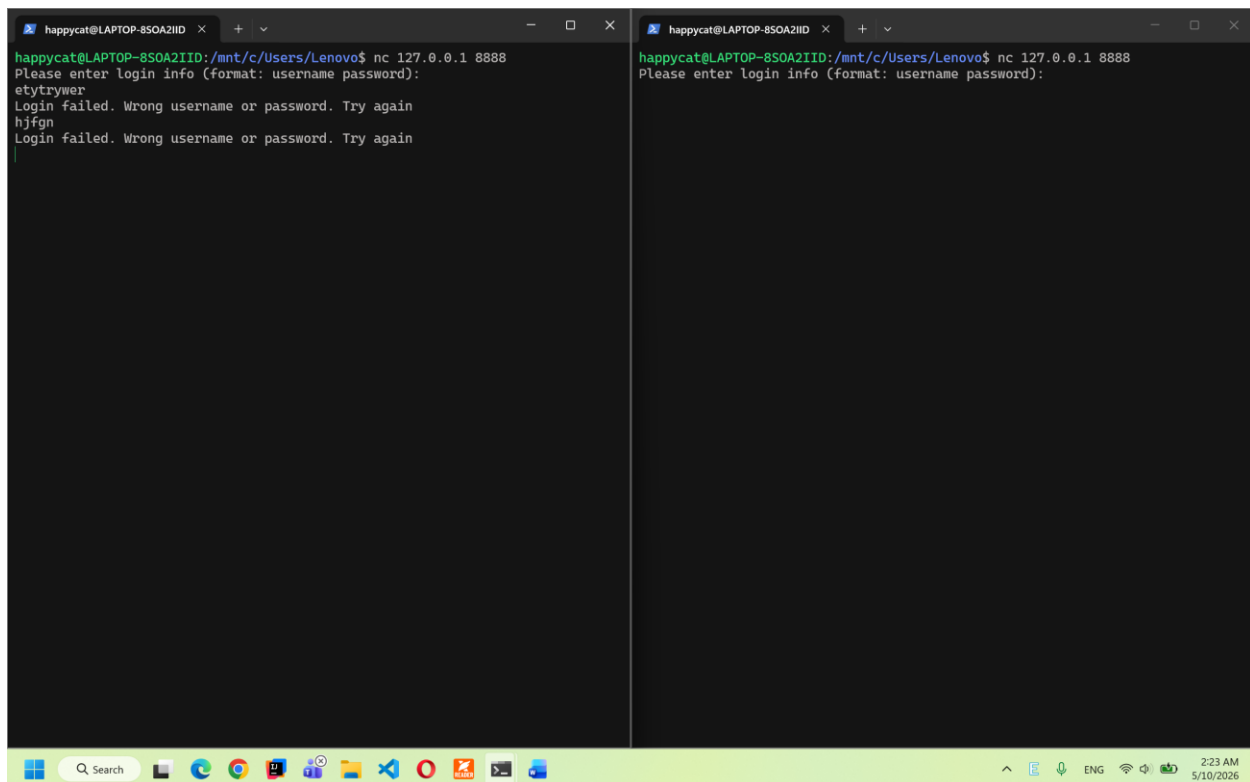
```
trinh_mang/network_programming_homework/hw_05_05$ cd "/mnt/d/VSCode/C vs code/lap...
trinh_mang/network_programming_homework/hw_05_05/" && mkdir -p build && gcc telnet_server_poll.c -o build/telnet_server_poll && ./build
/telnet_server_poll
Server is listening at port 8888...
```

A notification at the bottom right states: "Your application running on port 8888 is available. See all forwarded ports" with buttons for "Open in Browser" and "Preview in Editor".

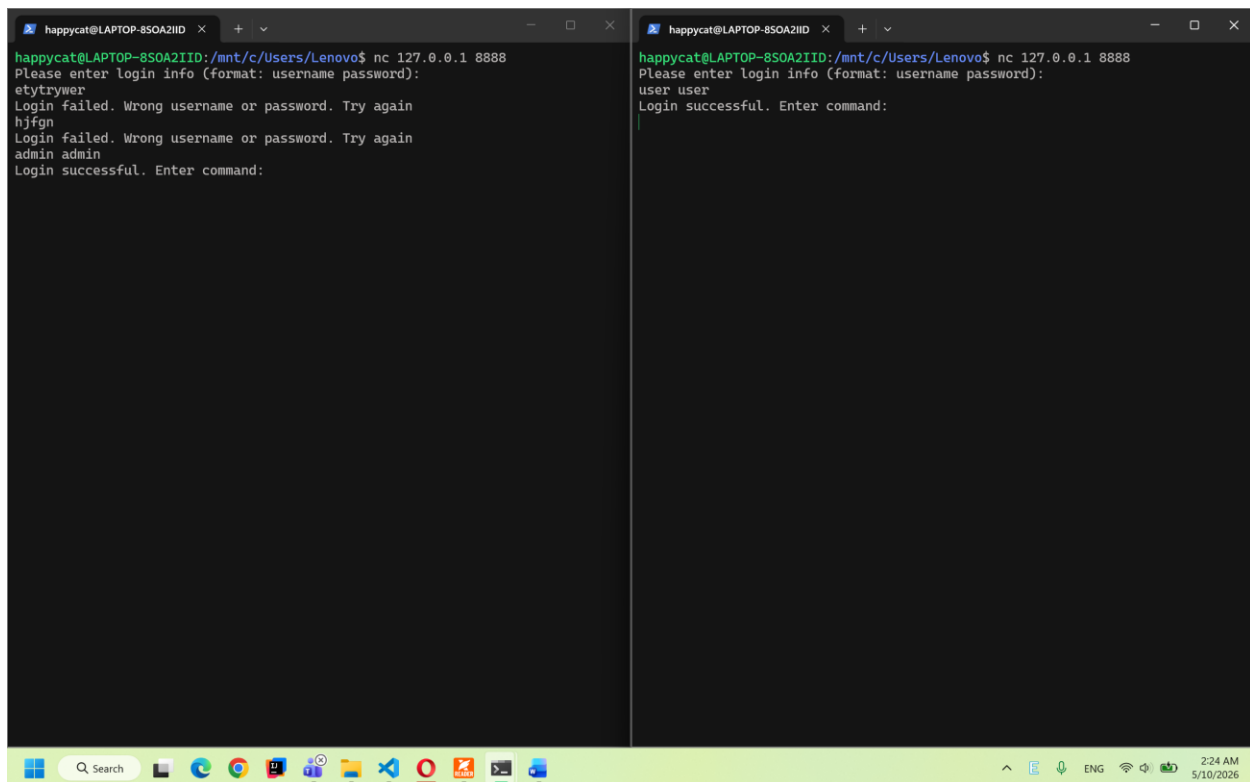
- Chạy 2 client kết nối vào server



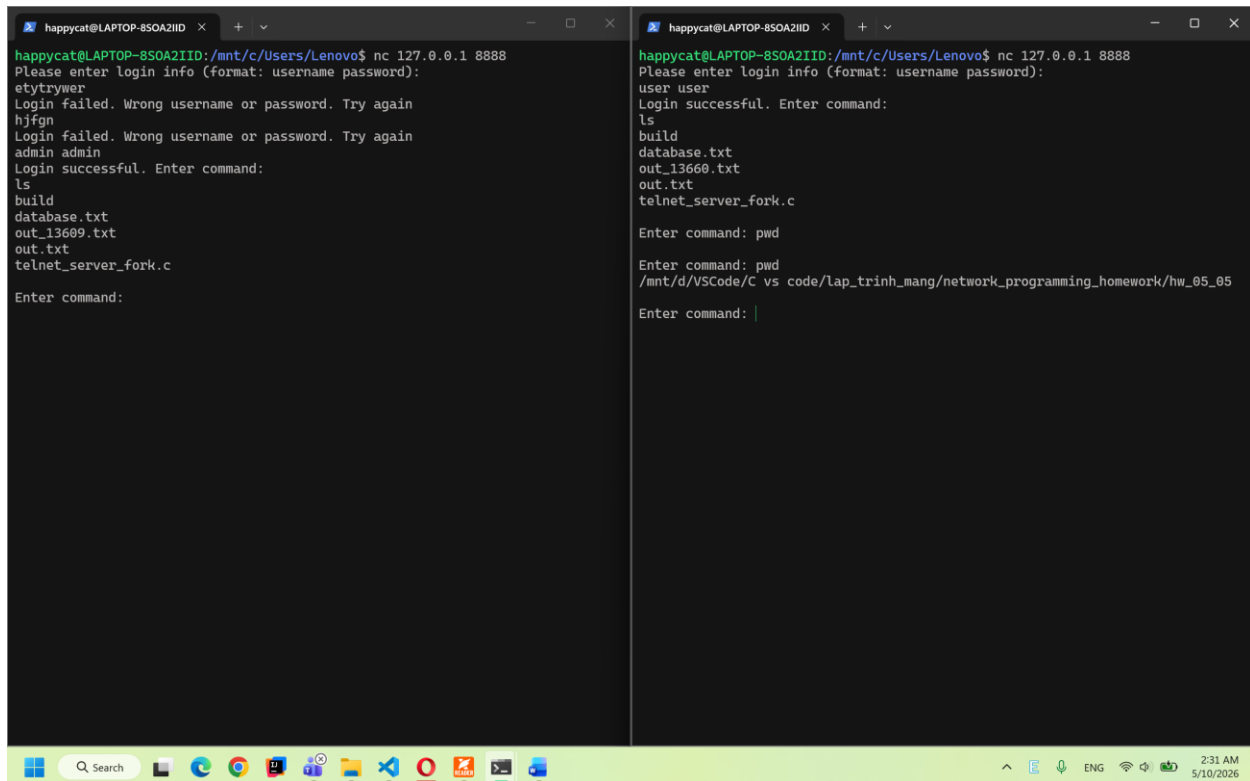
- Nhập username và password (format: “username password”)
- Nếu nhập sai, server sẽ báo lỗi và yêu cầu nhập lại



- Nhập đúng thì server sẽ báo thành công và yêu cầu nhập lệnh



- Nhập lệnh xong, server sẽ thực hiện, ghi kết quả ra file out_{pid}.txt và gửi về client, sau đó xóa file out_{pid}.txt đi



```
happycat@LAPTOP-85QA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Please enter login info (format: username password):
etytrywer
Login failed. Wrong username or password. Try again
hjfgn
Login failed. Wrong username or password. Try again
admin admin
Login successful. Enter command:
ls
build
database.txt
out_13609.txt
out.txt
telnet_server_fork.c
Enter command:

happycat@LAPTOP-85QA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Please enter login info (format: username password):
user user
Login successful. Enter command:
ls
build
database.txt
out_13660.txt
out.txt
telnet_server_fork.c
Enter command: pwd
Enter command: pwd
/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_05_05
Enter command: |
```

- Kết quả ở server

```
File Edit Selection View Go Run Terminal Help
network_programming_homework [WSL: Debian]

EXPLORER
OPEN EDITORS
telnet_server_fork.c hw_05_05 U
out.txt hw_05_05 U
database.txt hw_05_05 U
telnet_server.c hw_07_04
database.txt hw_07_04

NETWORK_PROGRAMMING_HOMWORK [...]
.vscode
chw_28_04
build
bai1.c
bai2.c
chw1
chw2
hw_05_05
build
database.txt
out.txt
telnet_server_fork.c
hw_07_04
build
baocao_hw-07-04_PhanMinhVuo...
chat_server.c
database.txt
telnet_server.c
hw_14_04
hw_28_04
hw1
hw2
.gitignore

OUTLINE
TIMELINE

telnet_server_fork.c hw_05_05 U
main()
// neu co yeu cau ket noi
// chap nhan va ghi vao mang clients
int client = accept(listen_sock, NULL, NULL);
printf("New client accepted: %d\n", client);

if (fork() == 0) {
    close(listen_sock); // dong listen_sock vi khong dung
    signal(SIGCHLD, SIG_DFL); // tien trinh con khong quan tam SIGCHLD

    bool client_check = false;

    // hoi user va pass
    send(client, ask_login, strlen(ask_login), 0);

    while (1) {
        // kiem tra nhan du lieu tu client
        int bytes_received =

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_05_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_05_05/" && mkdir -p build && gcc telnet_server_fork.c -o build/telnet_server_fork && ./build/telnet_server_fork
Server is listening at port 8888...
New client accepted: 4
Client 4 logged in successfully
Client 4 executor: ls
Client 4 executor: ls
Client 4 executor: pwd
sh: 1: pwd: not found
Client 4 executor: pwd

Ln 95, Col 19 (6 selected) Spaces: 4 UTF-8 LF {} C Linux
2:31 AM 5/10/2026
```

2. Bài 2 – http_server

- Chạy server ở cổng 8888

```
File Edit Selection View Go Run Terminal Help
network_programming_homework [WSL: Debian]

EXPLORER
OPEN EDITORS
telnet_server_fork.c hw_05_05 U
http_server_preforking.c hw_05_05 U
database.txt hw_05_05 U
telnet_server.c hw_07_04
database.txt hw_07_04

NETWORK_PROGRAMMING_HOMWORK [...]
.vscode
chw_28_04
build
bai1.c
bai2.c
chw1
chw2
hw_05_05
build
database.txt
http_server_preforking.c
telnet_server_fork.c
hw_07_04
build
baocao_hw-07-04_PhanMinhVuo...
chat_server.c
database.txt
telnet_server.c
hw_14_04
hw_28_04
hw1
hw2
.gitignore

OUTLINE
TIMELINE

http_server_preforking.c hw_05_05 U
main()
exit(0);

printf("Press Enter to exit...\n");
getchar();

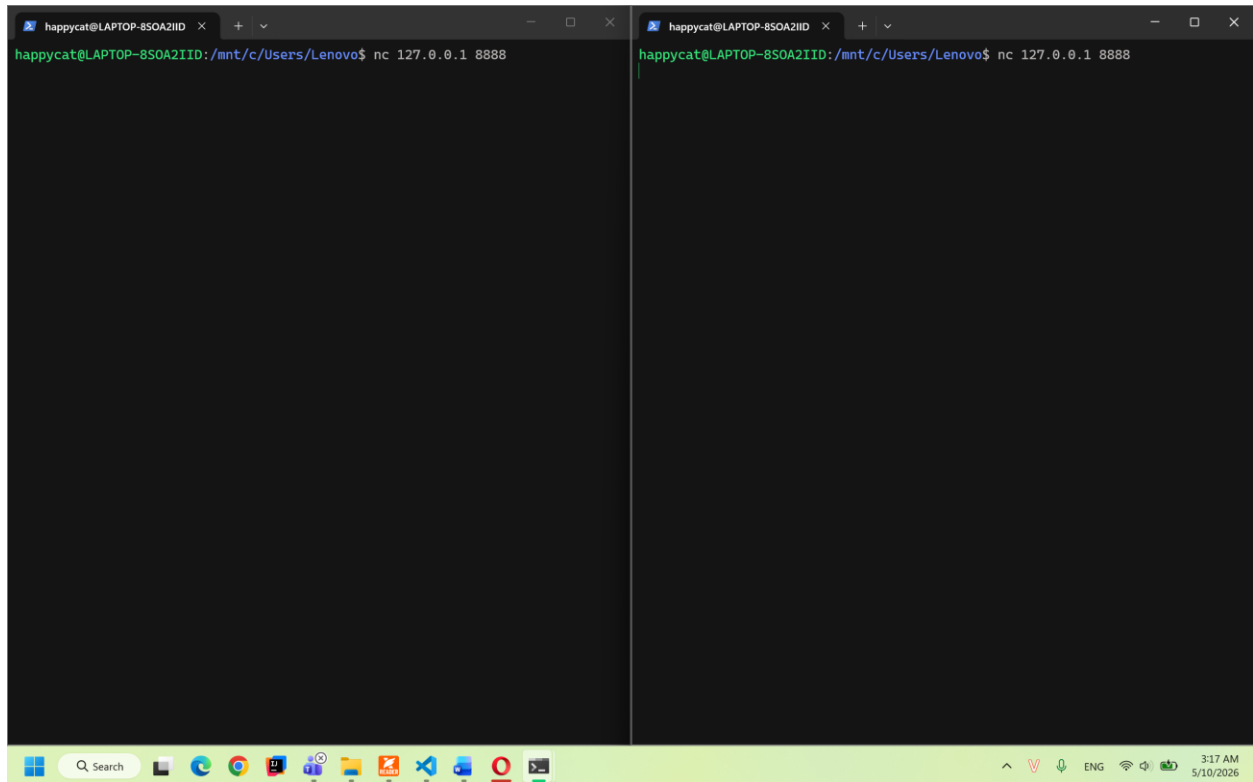
killpg(0, SIGKILL); // kill het tat ca tien trinh

close(listen_sock);
return 0;

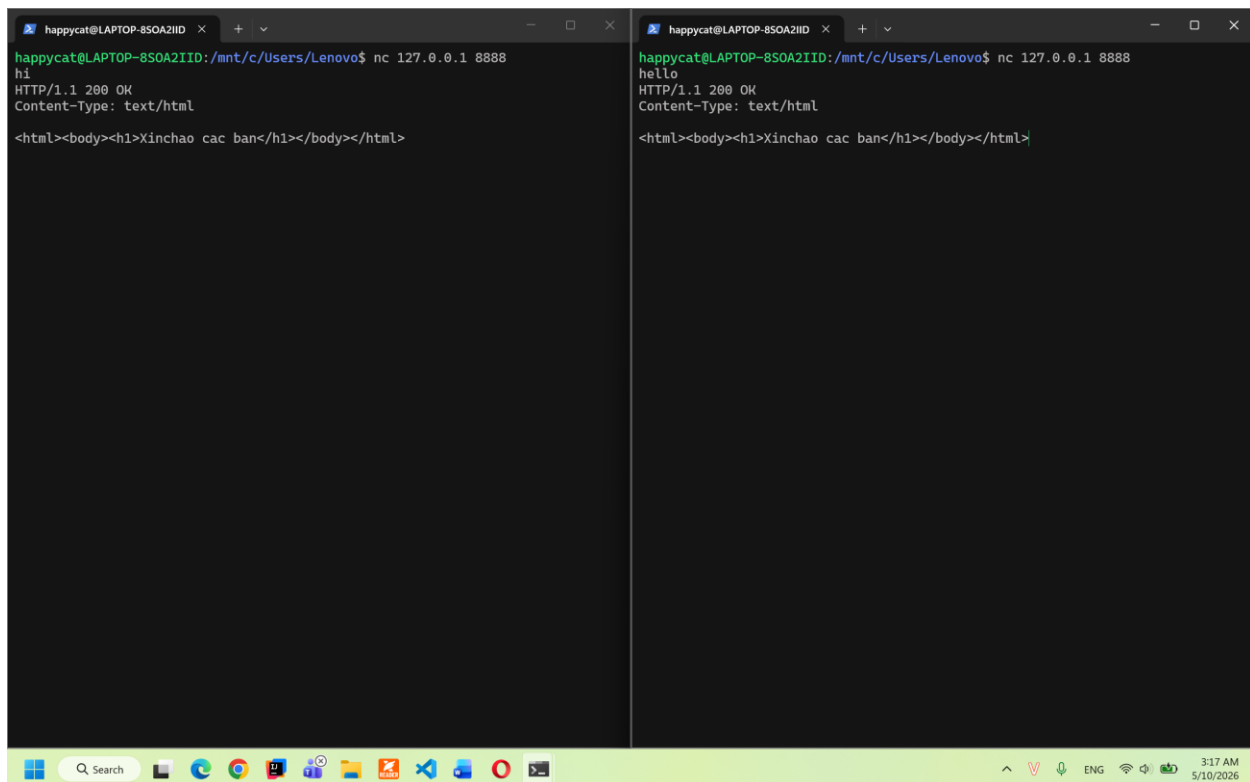
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_05_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_05_05/" && mkdir -p build && gcc http_server_preforking.c -o build/http_server_preforking && ./build/http_server_preforking
Server is listening at port 8888...
Press Enter to exit...

Ln 94, Col 24 Spaces: 4 UTF-8 LF {} C Linux
3:16 AM 5/10/2026
```

- Tạo 2 client kết nối với server



- Client gửi message và nhận được mã HTML từ server, sau đó server đóng kết nối với client vừa gửi



The image shows two terminal windows side-by-side. The left window is a web server (http_server_preforking.c) listening on port 8888. It receives a request from 127.0.0.1 and responds with an HTTP 200 OK status and a Content-Type of text/html. The response body is an HTML document with a single line: "Xinchao cac ban". The right window is a netcat client (nc) connected to 127.0.0.1 on port 8888. It sends a "hello" message and receives the same HTML response from the server.

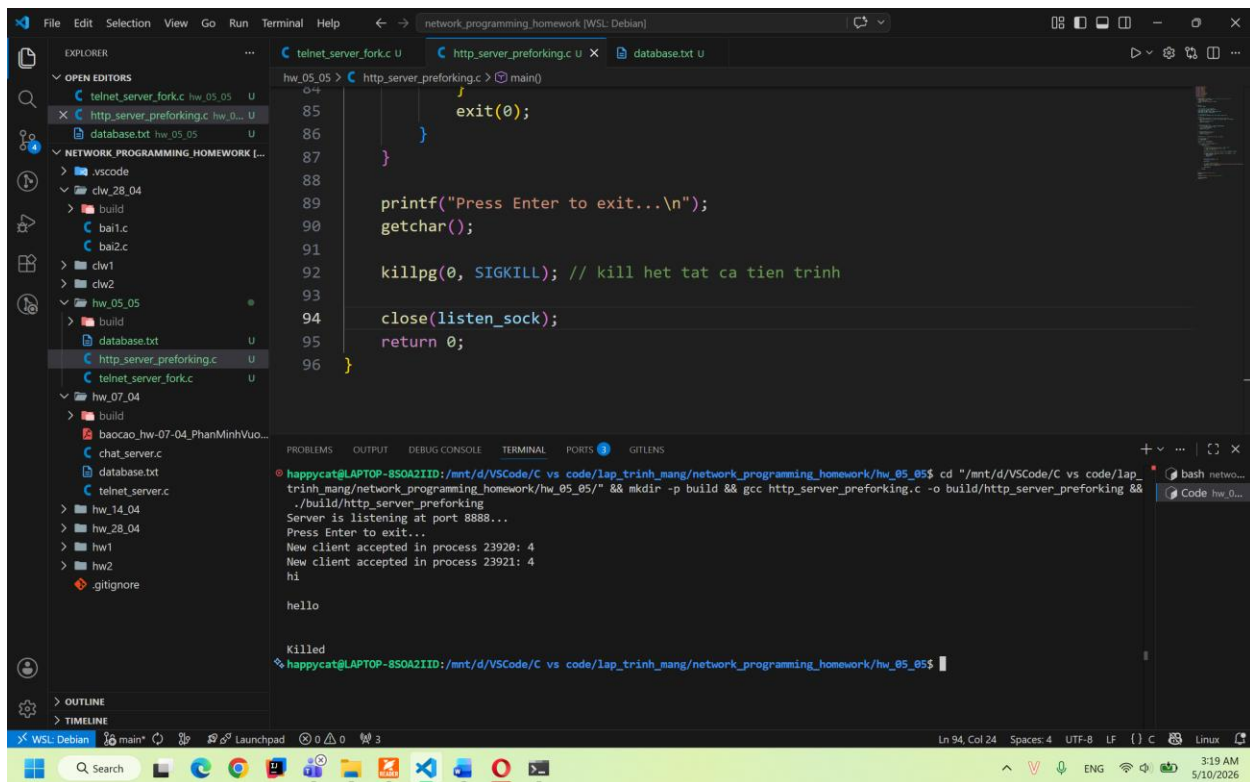
```
happycat@LAPTOP-85OA2IID /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
hi
HTTP/1.1 200 OK
Content-Type: text/html

<html><body><h1>Xinchao cac ban</h1></body></html>
```

```
happycat@LAPTOP-85OA2IID /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
hello
HTTP/1.1 200 OK
Content-Type: text/html

<html><body><h1>Xinchao cac ban</h1></body></html>
```

- Bên phía server, khi nhấn Enter sẽ kill tất cả tiến trình



The image shows a VS Code editor window with the file explorer on the left and the editor on the right. The editor is open to the file http_server_preforking.c in the directory /mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_05_05. The code in the editor is as follows:

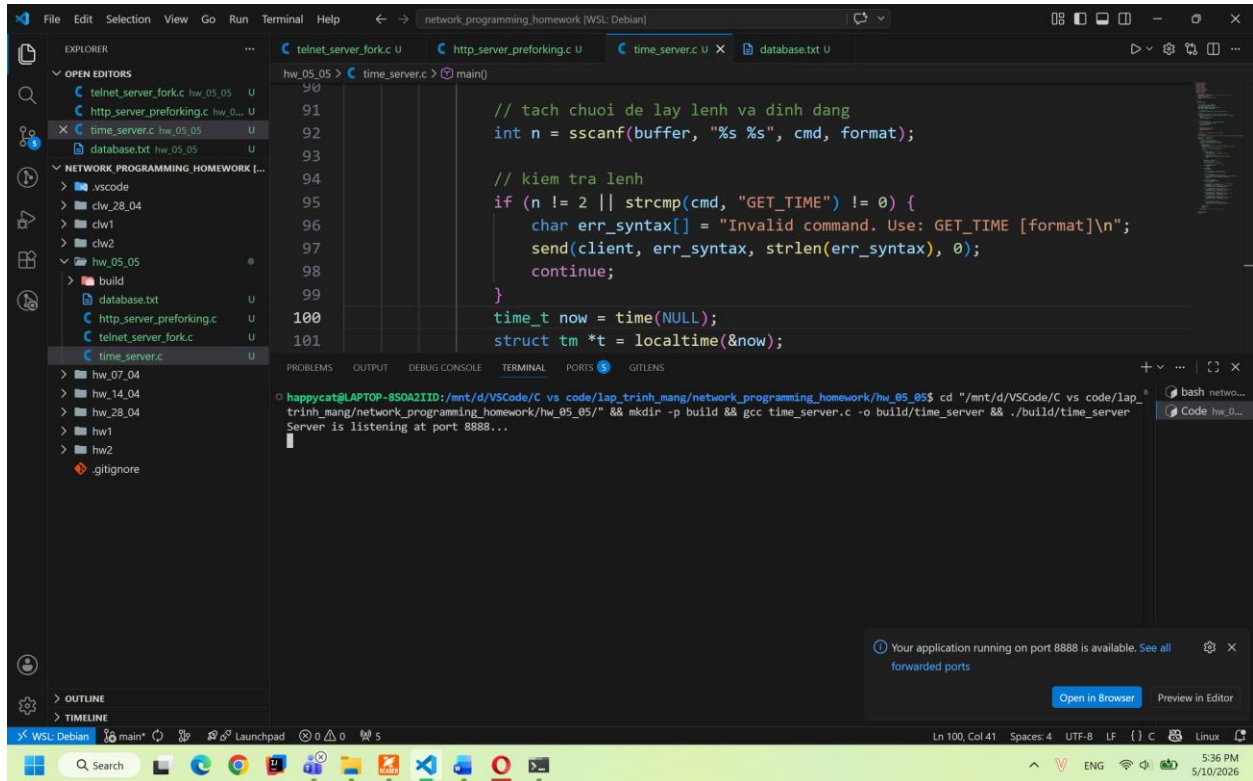
```
85     exit(0);
86 }
87
88
89 printf("Press Enter to exit...\n");
90 getchar();
91
92 killpg(0, SIGKILL); // kill het tat ca tien trinh
93
94 close(listen_sock);
95 return 0;
96 }
```

The terminal at the bottom shows the output of the program. It starts by listening on port 8888. It receives two connections from 127.0.0.1. The first connection receives the message "hi" and the second connection receives the message "hello". After the second connection, the program prints "Killed" and the terminal prompt returns to the user.

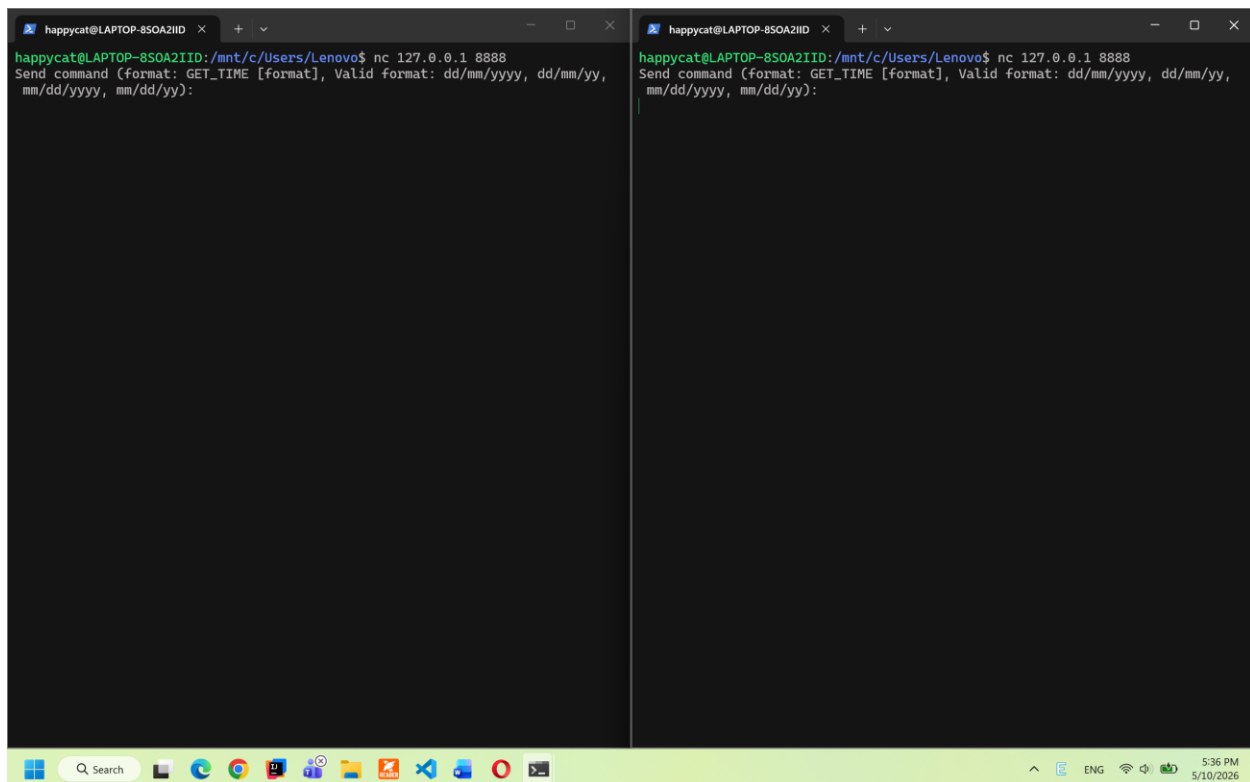
```
happycat@LAPTOP-85OA2IID /mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_05_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_05_05/" && mkdir -p build && gcc http_server_preforking.c -o build/http_server_preforking && ./build/http_server_preforking
Server is listening at port 8888...
Press Enter to exit...
New client accepted in process 23920: 4
hi
New client accepted in process 23921: 4
hello
Killed
happycat@LAPTOP-85OA2IID /mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_05_05$
```

3. Bài 3 – time_server

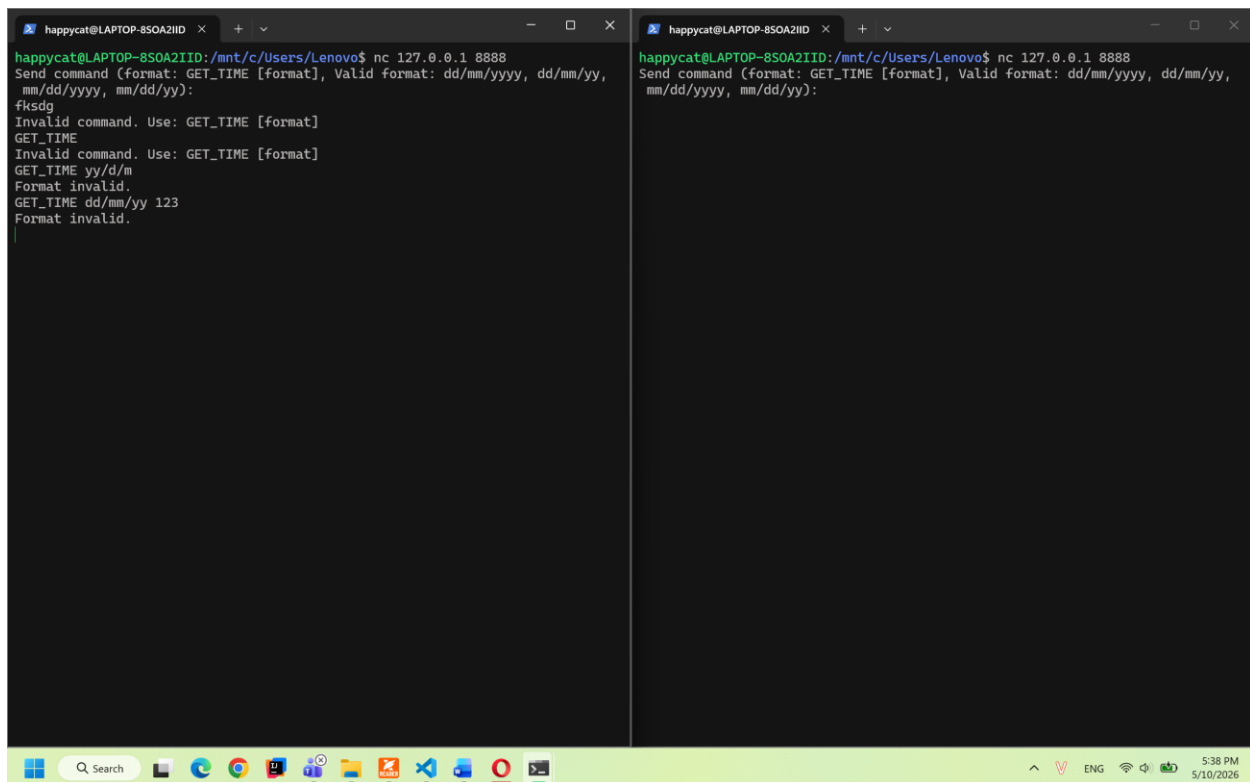
- Chạy server ở cổng 8888



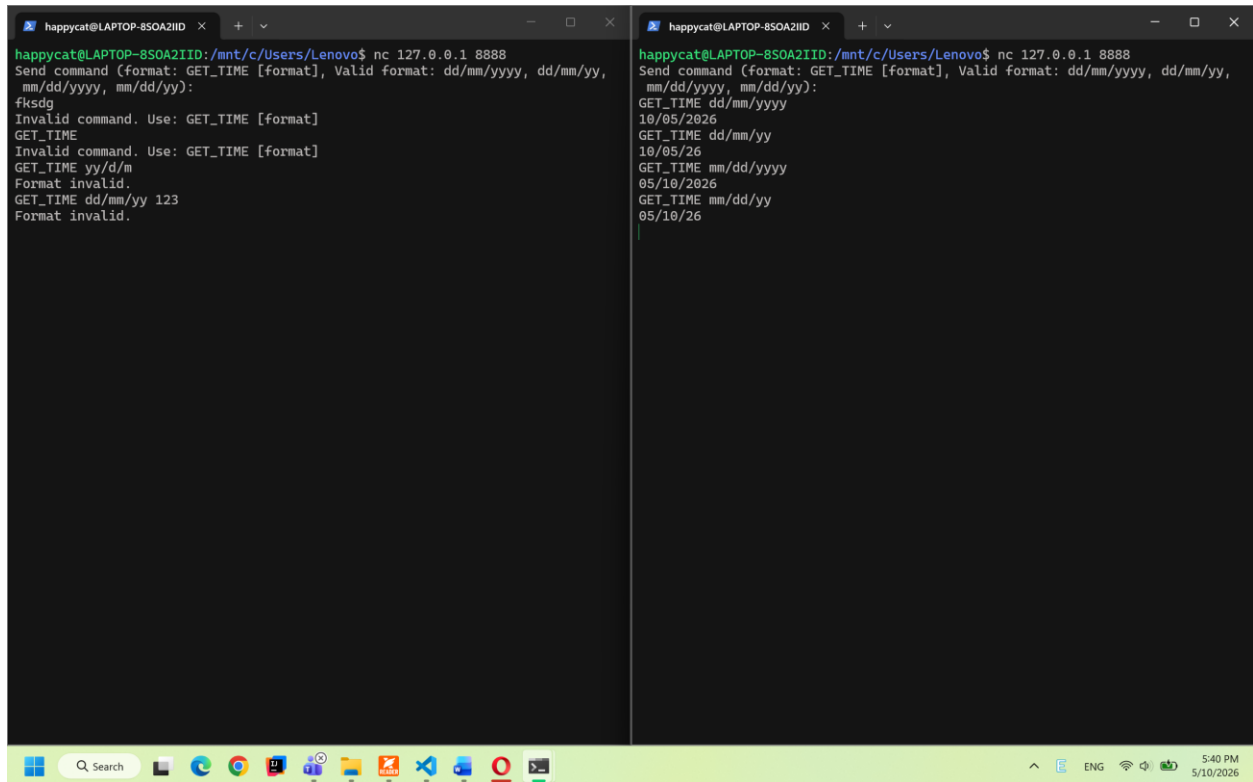
- Tạo 2 client kết nối với server



- Nếu nhập sai cú pháp, server sẽ yêu cầu nhập lại (thừa tham số, sai định dạng)



- Nhập đúng định dạng thì server sẽ gửi thời gian cho client



```
happycat@LAPTOP-850A2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Send command (format: GET_TIME [format], Valid format: dd/mm/yyyy, dd/mm/yy,
mm/dd/yyyy, mm/dd/yy):
fksdg
Invalid command. Use: GET_TIME [format]
GET_TIME
Invalid command. Use: GET_TIME [format]
GET_TIME yy/d/m
Format invalid.
GET_TIME dd/mm/yy 123
Format invalid.

happycat@LAPTOP-850A2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Send command (format: GET_TIME [format], Valid format: dd/mm/yyyy, dd/mm/yy,
mm/dd/yyyy, mm/dd/yy):
GET_TIME dd/mm/yyyy
10/05/2026
GET_TIME dd/mm/yy
10/05/26
GET_TIME mm/dd/yyyy
05/10/2026
GET_TIME mm/dd/yy
05/10/26
```